

Figure 4: Applicability-boundary summary
 (Green = identifiable, Yellow = conditional, Red = inconclusive/boundary)

Dataset / Model	AU Saturation	TwoNN Consistency	Seed Stability	Diagnostic Decision
MNIST FC-VAE	Yes	Yes ($\sigma \leq 0.12$)	Yes ($\sigma \leq 0.5$)	Identifiable
MNIST Conv-VAE ($\beta=4$)	No (β too low)	Partial ($m=64$ only)	Yes	Inconclusive
MNIST Conv-VAE ($\beta \geq 10$)	Yes ($m_{knee} \approx 128$)	Partial	Partial	Identifiable (high β)
CIFAR-10 (Conv-VAE)	Weak/conditional ($\beta=80$, thin dec.)	Partial ($m \approx 2d$)	No	Boundary
SVHN (Conv-VAE)	Partial	No (diverging)	No	Unreliable